

Profile

Staff Software Engineer with 13+ years of experience designing, stabilising, and delivering complex production systems across desktop, console, XR, mobile, embedded, and cloud platforms.

Specialised in recovering and modernising legacy or fragile systems, restoring build and delivery pipelines, and creating tooling and infrastructure that enable teams to ship reliably. Often brought into projects under technical or production pressure, I focus on understanding systems quickly, reducing complexity, and enabling sustainable delivery.

My experience spans rapid prototypes and MVPs through to live production releases, with particular strength in reverse-engineering undocumented systems, designing APIs and services, and resolving critical engineering challenges.

Technical Skills

Languages: C++, C#, TypeScript/JavaScript, Python, Rust, Java, Objective-C, Swift

Backend & Systems: API design (REST, GraphQL), Node.js/Express services, realtime networking (UDP, WebSockets, Photon), distributed systems, authentication systems, service integration, event-driven architectures

Platforms: Unity, Godot, Android (SDK/NDK), Windows, Linux, Steam, XR platforms

Cloud & Infrastructure: AWS (EC2, S3, RDS), Firebase, Google Cloud, Docker, CI/CD pipelines, build & release engineering, monitoring and production incident recovery

Professional Experience

Independent	UK, Remote
Senior Software Engineer (Freelance)	Jan 2024 – Feb 2026

- Engaged across multiple contracts to recover and modernise legacy production systems, restoring broken build pipelines, updating SDKs, and re-establishing reliable build environments.
- Reverse-engineered and reconstructed the core foundations of a 22-year-old arcade title on a modern engine, faithfully recreating original gameplay behaviour, timings, and data formats while enabling testing, modding, and future development.
- Built internal tooling to support designer-led iteration on an unreleased title, resolving long-standing technical debt from prior development cycles.
- Stabilised fragile codebases under tight time and budget constraints, bringing non-building projects back to a shippable state.

Virtual Realms	Bangkok, Thailand
Lead Software Engineer	Jan 2022 – Dec 2023

- Led development of a new MMORPG built on an existing codebase, owning both client and server systems from rapid prototyping through alpha delivery.
- Built internal tools and backend integrations to unblock production, including R&D for blockchain-based character systems when third-party solutions failed.
- Recovered and secured live production infrastructure after a database breach on the studio’s existing MMO.

Mooshoo Labs
Co-Founder & Senior Software Engineer

Gateshead, UK
Jan 2013 – Dec 2022

Co-founded a game development studio combining client consultancy with in-house product development across PC, mobile, and console. I specialised in client-facing consultancy, unblocking production through R&D, tooling, and stabilising legacy or time-critical codebases.

- Built and stabilised production systems across games, mobile apps, XR projects, embedded systems (medical devices), and startup MVPs; frequently brought in to resolve critical issues and meet hard deadlines.
- Specialised in rapid R&D, legacy system recovery, internal tooling, and unblocking technical challenges.
- Built in-house products end-to-end, from early prototypes through commercial release and successful exits.

General Interactive Co
Lead Programmer (Consultant)

Singapore, Remote
2020 – 2021

- Joined mid-production to stabilise and scale development of Steam based game within a multidisciplinary team.
- Scoped, implemented, and shipped new gameplay systems and missions on an existing codebase, improving robustness and reusability while resolving production-blocking bugs.
- Provided technical leadership through onboarding, implementation guidance, and delivery unblocking.

OLO Technologies
Consultant

Singapore, Remote
2020 – 2020

- Stabilised a production AR configurator under deadline pressure for enterprise OEM clients.
- Re-architected the data model and migrated the platform to cloud-backed infrastructure using Firebase, enabling multi-project, multi-device access.
- Owned client and backend systems, including API design, hosting and authentication, enabling successful delivery.

Personal Projects

Social Multiplayer Mobile App (iOS / Android)

In Development

Single-handedly designing and developing a cross-platform social multiplayer application enabling asynchronous turn-based gameplay between remote users.

- Designing backend services and APIs supporting authentication, session management, and synchronised gameplay across distributed clients (Node.js, GraphQL)
- Developing cross-platform client architecture and interaction systems supporting media capture, upload, and asynchronous gameplay workflows

Community Volunteering Platform

In Development

Designing and developing a full stack web platform helping users discover nearby volunteering events and community initiatives.

- Building responsive web interfaces and participation flows supporting event discovery and user engagement.
- Designing data storage, authentication flows, and external integrations including maps services.

Education

Teesside University
Bsc (Hons) Computer Games Programming 2:1

Middlesbrough, UK
2009 – 2011